

Mohamed Rayan Zaarir

Full-Stack Engineer

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SUMMARY

Full-Stack Developer with 3+ years of experience in the tech industry, specializing in backend engineering and scalable system design. Experienced in building end-to-end applications using Django, Django REST Framework, React, and Next.js, with a strong emphasis on robust backend architecture.

Proven track record in architecting high-complexity platforms involving advanced financial workflows, inventory traceability, and multi-role access control. Strong background in PostgreSQL schema design, distributed task processing (Celery/Redis), and production deployment using Docker, Nginx, Gunicorn, and CI/CD pipelines. Passionate about clean architecture, performance optimization, and long-term maintainability.

SKILLS

Backend	Django, Django REST Framework, FastAPI, JWT Auth, Role-Based Access Control
Frontend	React.js, Next.js, TypeScript, JavaScript (ES6+), Tailwind CSS, shadcn/ui
Database & Cache	PostgreSQL (Schema Design, Query Optimization), Redis, Data Modeling
Async Processing	Celery, Celery Beat, Background Task Queues
DevOps	Docker, Docker Compose, Nginx, Gunicorn, GitLab CI/CD, DigitalOcean, Linux
Architecture	Clean Architecture, SOLID Principles, OOP, Scalable System Design
Languages	Arabic (Native), English (Professional), French (Professional)

EXPERIENCE

Backend Engineer August 2025 – Present

Originova – Client Project: Yassir Distribution Management System

- Architected and delivered a large-scale distribution and inventory management platform supporting up to 1000 concurrent users across five operational roles.
- Designed and optimized a highly normalized PostgreSQL schema with 150+ relational tables covering inventory control, financial accounting, supplier/client ledgers, rentals, penalties, and audit logging.
- Engineered 500+ RESTful API endpoints powering desktop and mobile applications.
- Implemented complex financial workflows including credit management, debt tracking, automated ledger entries, and payment reconciliation systems.
- Integrated batch-level inventory traceability with expiry monitoring, stock transfers, and automated alert mechanisms.
- Developed asynchronous processing pipelines using Redis, Celery, and Celery Beat for scheduled jobs, financial calculations, and background tasks.
- Containerized and deployed the system on DigitalOcean using Docker, Nginx (reverse proxy), Gunicorn (WSGI), SSL configuration, and GitLab CI/CD pipelines.

Full-Stack Engineer (Freelance) October 2025 – December 2025

Lift & Cable Installation Company Management Platform

- Architected and delivered a full enterprise management system using Django REST Framework, React, TypeScript, and Tailwind CSS.
- Implemented complete project lifecycle management including client registration, project creation, verification workflows, employer assignment, and maintenance scheduling.
- Designed advanced inventory management with real-time stock tracking, reorder alerts, and profit margin calculations.
- Developed a flexible invoice management system with multi-line invoicing, product integration, and financial tracking.
- Built an interactive calendar system for project scheduling and maintenance planning with advanced filtering capabilities.
- Secured the platform using JWT authentication and role-based access control.
- Implemented a comprehensive analytics dashboard with financial and operational insights.

Full-Stack Engineer March 2025 – June 2025

Final Year Project Management Platform (University System)

- Designed and implemented end-to-end application features managing 60+ projects and 200+ users.

- Built backend services using Django and DRF alongside frontend interfaces for project lifecycle management.
- Implemented real-time collaboration features using Django Channels and Redis.
- Delivered a modular, containerized architecture using Docker.
- Produced comprehensive API documentation with DRF-YASG.

Full-Stack Engineer

January 2025 – March 2025

Official State Library of Tisemsilt

- Engineered a full-stack platform serving thousands of monthly users.
- Integrated PMB library system via JRPC web services.
- Improved performance by 40% using Redis caching.
- Delivered scalable infrastructure using Django, PostgreSQL, Redis, and Docker.

Full-Stack Engineer (Mobile & Backend)

January 2024 – June 2024

Lotok – Car Rental Platform

- Developed backend services for authentication, booking, and car management.
- Built Android application using Kotlin and Jetpack Compose.
- Integrated Google and Facebook authentication.
- Ensured seamless frontend-backend integration via containerized services.

Network Engineering Intern

July 2025 – August 2025

Algérie Télécom

- Worked on enterprise networking infrastructure including routing, switching, LAN/WAN configuration, and firewall basics.
- Participated in diagnosing multi-site connectivity issues.

EDUCATION

Master's degree in Computer Science

2027

Higher National School of Computer Science (ESI-SBA), Sidi Bel Abbès

- Software Engineering and System Design specialization.
- Focus on distributed systems, algorithms, and scalable architectures.

Baccalaureate

2022

Kritli Mokhtar High School, Blida

- Graduated first in the city with distinction.